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Automatic building footprint extraction from very high-resolution imagery using deep learning techniques

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ABSTRACT

Building footprint maps are useful for urban planning, infrastructural development, population estimation and disaster management. With the availability of very-high resolution satellite imagery, remote sensing community is pursuing automatic techniques for extracting building footprints for cities with varried building types. Recently, CNNs (Convolutional Neural Network) have been successfully applied for extraction of building footprint from satellite imagery. In this paper, we propose a novel CNN architecture termed UNet-AP inspired by UNet and the concept of Atrous Spatial Pyramid Pooling, for automatic extraction of building footprint from very-high resolution satellite imagery. We demonstrate extraction of building footprints from Cartosat-2 series 4-band (Blue, Green, Red and Near-Infrared) multispectral satellite imagery, pan-sharpened using the panchromatic image with less than 1-meter resolution. We also compare the performance of our proposed architecture with baseline implementation of recently proposed UNet and SegNet architectures. We present a comparative assessment of the architecture performance across different types of urban settlement classes such as dense built-up areas, slums and isolated buildings. We demonstrate that our proposed architecture outperforms SegNet and UNet in terms of overall mean intersection over union (0.75 vs 0.70 and 0.68 for UNet and SegNet respectively) and delivers consistent improvement across all three settlement classes.

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KEYWORDS
Convolutional Neural Network; SegNet; UNet; UNet-AP

1. Introduction

Very high-resolution satellite imagery is one of the primary sources for updating and validating building footprint maps. These building footprint maps are crucial at the time of disaster, urban planning, infrastructure development, population estimation, 3 D city modelling and other geospatial related applications (Jensen and Cowen 1999). Manually delineating buildings by visual interpretation for large areas is a laborious, expensive and time-consuming task and there is a huge requirement for developing methodologies for automatically extracting building footprint.

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In the past, many methods based on automatic or semi-automatic processes were developed for building footprint extraction. The large variations in appearance, geometry and spectral properties of buildings, altogether makes it a challenging task to enable automated building extraction at large scale from remote sensing satellite imagery. As per the spectral, geometrical properties of buildings such as rectangular shape, homogeneous surface (uniform spectra) and low-level features such as edges, lines/corners and association of shadows with the buildings are few of the fundamental element of buildings (Lnglada 2007).

In recent years, many tools for data collaboration are used for updating building footprint maps, following the concept of 'crowdsourcing'- where large groups of users can perform functions that are either difficult to automate and implement (Haklay and Weber 2008), is used for updating building footprint maps. The increasingly popular Volunteered Geographic Information (VGI) from OpenStreetMap (OSM) might be considered as a source for obtaining large scale building maps, where manual labelling of building footprints is available. However, the inconsistent quality of VGI with poor positional accuracy, makes it inappropriate for use and requires large manual efforts for correcting the data.

Recently, Convolutional Neural Networks (CNNs) have been successfully used for extracting information from satellite imagery. We propose a novel CNN architecture based on UNet and Atrous Spatial Pyramid Pooling (ASPP) concept, termed as UNet-AP. The comparison of performance of our proposed architecture is with the baseline implementation of recently proposed SegNet and UNet architectures. Although these architectures are widely accepted, we would like to improve the existing models for delineating building boundaries, in order to make an end-to end architecture. Multi-source data such as height information obtained from Digital Elevation Model (DEM) and Light Detection and Ranging (LIDAR) is important information for extraction of building footprint maps. However, due to limited availability of high-quality DEM and LIDAR datasets for large area, we demonstrate extraction of building footprints from Cartosat-2 series 4-band (Red, Green, Blue and Near-Infrared) multispectral satellite imagery, pan-sharpened using the panchromatic image with less than 1-meter resolution with good accuracies. In this paper, we present a comparative assessment of the architectures for automatic extraction of building footprint as per the urban settlement classes such as dense built-up areas, slums and isolated buildings using deep learning techniques.

2. Related work

In remote sensing communities, a significant amount of work is done on building extraction from high resolution satellite imagery. Classification of building footprints from the satellite imagery can be categorized into two: pixel and object based. Pixel based classification of very-high spatial resolution data results in salt-and-pepper effect due to high inter-class spectral variability in an image (Li and Shao 2014). On the other hand, object-based image analysis overcomes this issue by segmenting image into meaningful objects as per the homogenous spectral values, considering factors such as the size of the object, texture, shape and context. Since segmented objects are the building-blocks for image in object-based classification, optimal segmentation is required considering large variation of size, shape and spectral properties of buildings in urban areas. Also, good feature engineering is required for better classification accuracy. Hence, the accuracy of classification depends on both features and parameters used for segmentation for optimal size of the object. The combination of spectral and geometrical properties of the object for extraction of building footprint have been used in Huertas and Nevatia (1988); Irvin and McKeown (1989); Guercke and Sester (2011). Morphological based filters such as Top-hat filter along with k-means algorithm is used by Gavankar and Ghosh (2018) and other methods such as graphs and energy minimizations models are used for correctly delineating building edges (Kim and Muller 1999; Tan et al. 2016). Also, height information from DEM obtained from optical stereo pair images using image matching techniques or LIDAR are used independently or in combination with the high-resolution images (Zhou and Neumann 2008; Awrangjeb et al. 2010). However, one of the limitations of DEMs generated is low accuracy and quality compared to LIDAR. The high-quality LIDAR data contributed in improving the delineation of building edges which otherwise contains noise (Wang et al. 2006).

The hand-crafted methodologies are successfully used for building footprints extraction; however, they are not robust enough to classify large variety of buildings based on their appearances and context. The development of CNNs and its combination with remote sensing images, increase the generalization capability of classification. CNN based integration of multi-layer and multi-sensor information for pixel-wise classification of building footprint is done by (Bittner et al. 2018; Yuan 2018). Fully convolutional neural network (FCN) and conditional random field as recurrent neural network used along with varied band combination such as (RGB) and (NIR, R, G), collectively used as an average, for the classification of building footprints (Yang et al. 2018). SegNet and ResNet architecture were used to implement multi-scale deep FCNs and early and later fusion of multi-modal remote sensing data such as LIDAR point cloud with multi-spectral data (Audebert et al. 2018). A modified UNet by adding batch normalization wrapper model along with numerous image enhancements is used for detection of building footprint (Prathap and Afanasyev 2018). Zhang et al. (2016) uses the pretrained AlexNet CNN for directly extracting remote sensing image features, and the extracted features are then used to train a classifier for urban land classes from google earth remote sensing images. Wu et al. (2018) proposes multi-constraint fully convolutional network (MC–FCN) model with bottom-up/top-down architecture is proposed for building segmentation through high-resolution aerial image, using optimized constrained intermediate layers. Alshehhi et al. (2017) uses a single patch-based Convolutional Neural Network (CNN) architecture for extraction of roads and buildings from high-resolution remote sensing data considering low-level features such as asymmetry and compactness of the objects in CNN during post-processing. Also, Xu et al. (2018) used the normalized differential vegetation index (NDVI), the normalized digital surface model (NDSM) and the first component of the principal component analysis (PCA), together with the original bands (R-G-B-IR), as the input from multi-source data has been used in CNN to extract buildings and vegetations. The result of these experiments proved that this approach could improve accuracy of extracting geographic objects in RS images,but due to less availability of multi-source data such as LIDAR and high resolution DEM and noisy DSMs which can influence the results largely and pose a problem with large area mapping. With huge availability of very high-resolution satellite data, good accuracy could be achieved using single source remote sensing multispectral data.

3. Study area and datasets

The study area is part of Ahmedabad city in Gujarat, India (Figure 1). The extent of the covered area lies between 22° 88′ 20″ N to 23° 16′ 20″ N latitude and 72° 48′ 20″ E to 72° 62′ 40″ E longitude, with an area of 594 sq. km. Ahmedabad is located on the banks

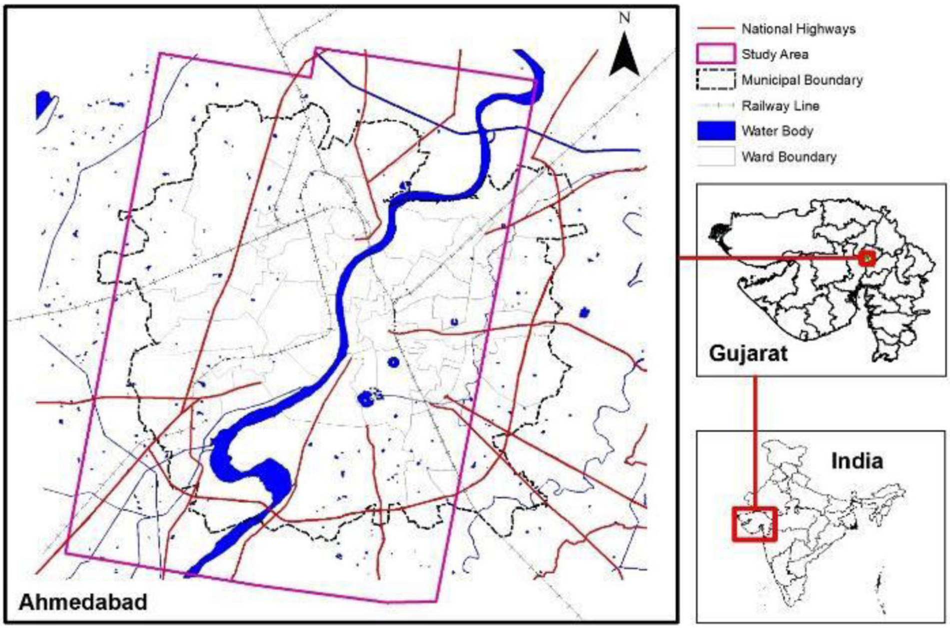


Figure 1. Study area for the part of Ahmedabad city.

of Sabarmati river. It emerges as an important economic and industrial hub in India. As per the AMC (2006), the inner city is considered as Central Business District (CBD) with dense building structures, the eastern bank of Sabarmati river has old and dense building structures with several industries and the western bank of Ahmedabad are predominantly having residential, commercial and isolated buildings around farm lands. The Indian remote sensing satellite Cartosat-2 series data acquired on October, 2018, with panchromatic band having less than 1 m ground spatial distance (GSD) and multispectral bands (blue, green, red and near-infrared) with 1.6 m GSD is used for extracting building footprints. The ortho-rectified, pan sharpened image using panchromatic and multispectral data, with less than 1 m GSD is used by removing any building occlusion.

4. Methodology

4.1. Convolutional neural network

Convolutional Neural Networks (CNNs) being one of the most used deep learning networks have huge achievement in image classification, object detection and image segmentation. For the most part, regular CNNs incorporate convolutional layers with a set of feature maps that are convolved with a large number of filters subjected for training and these are then passed through some non-linear activation function for creating new feature maps. These feature maps are down-sampled by pooling layers. Passing through various layers, higher level features get encoded and in the last stage, output feature maps have most discriminative features for representing image classification, the structure of standard CNN is shown in (Figure 2). Extracting building footprint from remote sensing imagery requires semantic segmentation, known as pixel-wise labelling for delineating complete boundaries of buildings.

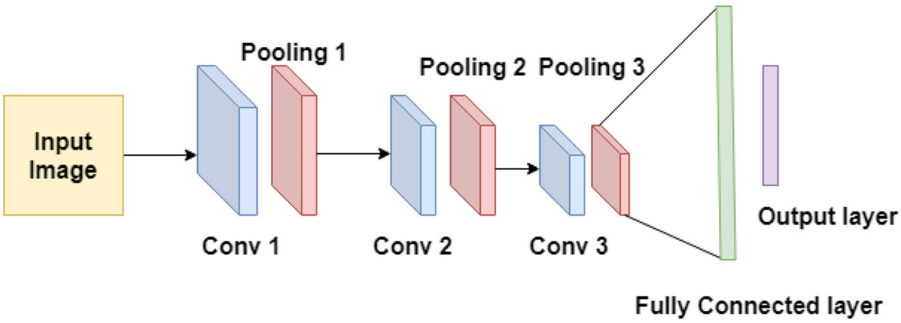


Figure 2. Architecture of Convolutional Neural Network, integrating multi-layer feature maps.

4.2. SegNet

SegNet model was first proposed by Badrinarayanan et al. (2017). The encoder-decoder architecture as in VGG16 (Simonyan and Zisserman 2014), consists of 13 convolutional layers of 3×3 convolutions and five layers of 2×2 max pooling. Decoder is a mirrored version of the encoder which uses the max-pooling indices of the encoder, for non-linear upsampling of the input feature maps followed by a pixel-wise classification layer. The model was developed to address the imperfect boundary delineation issues observed in fully convolutional networks and other similar networks for semantic image segmentation. The SegNet architecture is beneficial in our case when working with buildings, as it stores indices of max-pooled features which map strong responses from the convoluted inputs that usually responds to location of edges of the objects (Yang et al. 2018).

4.3. UNet

For image segmentation, one of the famous architectures used is UNet, it was originally developed for biomedical image segmentation (Ronneberger et al. 2015). Basically, the UNet builds upon the Fully Convolutional Network (Long et al. 2015). UNet model consists of an encoder layer (down sampling) for capturing the context in the image along with the max pooling layers. The second path of the model is decoder layer (up-sampling) for precise localization using transpose convolution. The limited availability of training data sets, encourages the adoption of UNet as an appropriate choice.

4.4. Atrous spatial pyramid pooling

The atrous convolution for dense feature extraction with variant field-of-view (FOV) is proposed by (Chen et al. 2018). In signal processing, atrous convolution is expressed by (Equation (1)),

$$y[i,j] = \sum_{k=1}^{K} x[i + r \cdot k, j + r \cdot k]\mathrm{w}[k] \tag{1}$$

Where $y[i,j]$ is the atrous convolution output, $x[i]$ in the input signal, w denotes the kernel of the convolutional filter with length k, and r denotes the rate parameter corresponding to the stride with which input parameter is sampled. Sampling rate denotes the FOV, which means higher the rate, higher the FOV. ASPP can implement multiple atrous convolutional layers in parallel manner with different sampling rate. Compared to regular convolution with large filters, atrous convolution allows to enlarge the FOV without

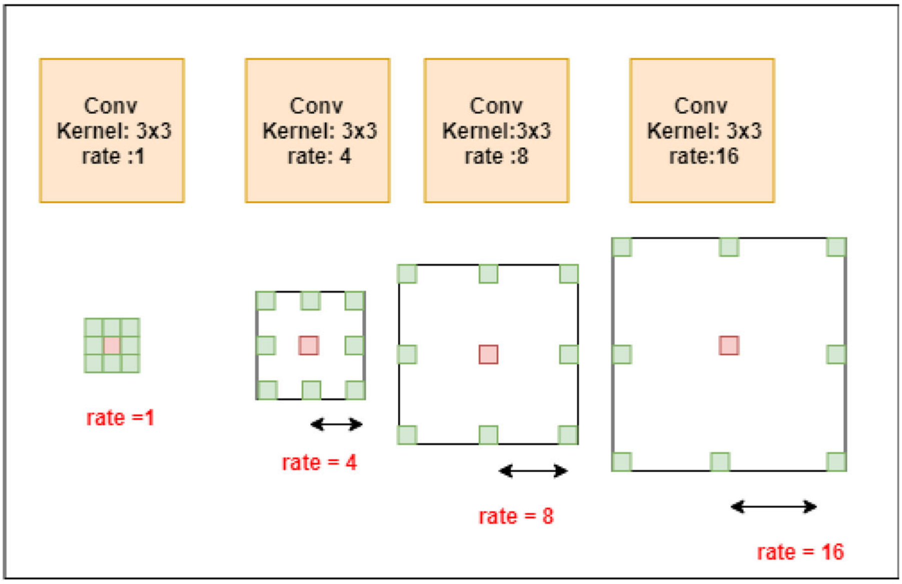


Figure 3. Atrous Spatial Pyramid Pooling (ASPP). Classifying the central red pixel by multiple parallel filters with different rates increasing the effective field-of-view.

increasing the parameters or the amount of computation (Chen et al. 2018). An illustration of ASPP with different rates is shown in (Figure 3).

4.5. UNet-AP

Using Atrous spatial pyramid pooling (ASPP) and UNet, we proposed the UNet-AP model taking the advantages of both networks inspired by (Chen et al. 2018). The additional contextual information is added in UNet through UNet-AP in the bottom layer. The ASPP technique is utilized in the bottom layer in order to incorporate multi-scale deep features into a discriminative feature in UNet. In the UNet-AP model as the original UNet network, each contracting path has a corresponding expansive path, allowing the contextual information to pass through the network. The contracting path consists of repeated convolution of 3×3 padded convolution, followed by non-linear activation of feature maps by exponential linear unit 'elu' activation, helping the cost function to smoothly converge to 0, it is given by (Equation (2))

$$f(x) = \ \{x \ for \ \left(x > 0\right) \ \& \ \alpha(expexp \ (x) \ -1) \ for \ (x < 0\Big) \tag{2}$$

where α is a hyperparameter with value in range (0,1) and followed by a 2×2 max-pooling operation with stride of size 2 for down-sampling.

At each down-sampling step a number of features are doubled and up-sampling of feature maps in the expansive path followed by a 2×2 convolution and two 3×3 convolutions followed by 'Elu'.

For optimising all the weights of the network efficiently, weights updating technique such as stochastic gradient descent (SGD) (Bottou 2010) optimization algorithms is used. L1-L2 regularization is used to penalize complexities and focuses on the most relevant features, and thus avoids overfitting of data (Demir-Kavuk et al. 2010). L1 is Lasso regression given by sum of absolute weights, assigning insignificant input feature with zero

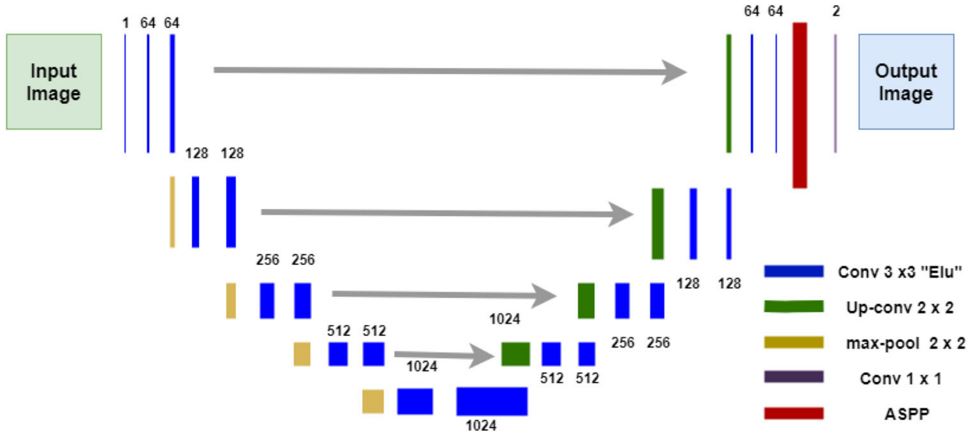


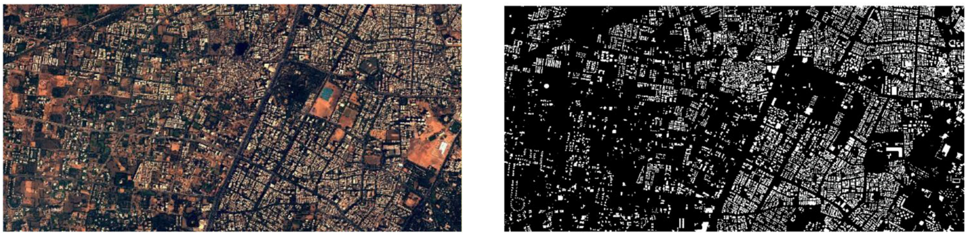
Figure 4. UNet-AP architecture, with blue layer corresponds to a multi-channel feature map. The number of channels is denoted along the box.

weight and L2 is known as ridge regression, which is the sum of square of all feature weights, assigning insignificant input feature with significantly low non-zero value, this particular regularizer is used when all the input feature plays vital role in output. A combination of L1 and L2-regularizer, which is the sum of the absolute and the squared weights is used in the model.

An atrous convolution with rates = 1, 4, 8 and 16 (Figure 4) are implemented in parallel before the softmax layer. The spatial resolution in the last layer is the same as the input image. Finally, a 1×1 convolution is applied to the map in the last upsampling layer to the class, and the softmax layer is applied to transform the feature to a probability distribution map belonging to each class having value in range (0,1).

5. Training data labelling and pre-processing

Like other supervised learning algorithms, CNN-based deep learning needs to learn features from a large number of training data to achieve a satisfactory generalized model. Training data from remote sensing images are very less available. However, training data for building footprint data available from crowdsourcing sites such as open street map (OSM) have poor positional accuracy with reference image, along with missing and erroneous building shapes. The available building footprint in OSM are in vector format (polygons), these are corrected according to building shape and georeferenced with reference to satellite image of the study area. Inaccuracies were corrected and vector data were converted into raster format with binary class as buildings with value 1 and non-buildings as 0. The example of building footprint dataset prepared is shown in (Figure 5) and which are used for model training. The training dataset is prepared from the study area itself, it comprises a heterogeneous set of buildings within urban areas including isolated buildings, dense built-up and slums. Isolated buildings are mostly in the periphery of the city, around the vacant lands and agricultural areas. The dense built-up areas cover the major part of the city in and around the central building district comprising commercial and residential areas. The slums are present within and outside the city with low spectral reflectance as compared to other building types and buildings are close to each other with distance less than 2 pixels making it difficult to delineate even with visual interpretation. In total fourteen thousand building footprints were prepared through visual interpretation



(a) (b)

Figure 5. (a) Satellite image of training data sets (b) labelled mask.

from the study area and out of which 80 percent is used for training and 20 percent is used as test data. The selection of these regions inside the study area depended on the grids of size 1 km × 1km overlayed on it. The grids with a heterogeneous blend of significantly all building types are included in the training and testing data. Other than these, grids dominated with a single class are also used for training and testing data. Majority of the grids are dominated by dense built-up types because these kinds of buildings are spread across the study areas.The idea behind creating a training and testing data set is to include all the building types with sufficient examples and are prepared across the study areas. Below in (Figure 5), is the sample of satellite image from the dataset, along with the corresponding mask. For incorporating sufficient robustness to the network and for creating large training datasets, we applied data augmentation techniques to our training datasets. The cartosat-2 imagery with four channels images (blue, green, red and NIR) along with the corresponding labels are used for training and testing the model. Also, the input training sets are rotated at (45°,90°,135°,180°) angles.

6. Experiments

For the task of semantic segmentation, a majority of work is done using RGB or color-infrared (CIR) images as input and most deep learning techniques developed by the computer vision community mainly deal with RGB images. In this study we have altogether used four bands red, green, blue and NIR for SegNet, UNet and UNet-AP network. The image patches are cropped in size of 264×264 pixels as the contextual information of the building is available within this size. The large and variant FOV influence the output with boundary artefacts in prediction. For preventing such artefacts and discontinuity in the boundary, we used an overlap of 64 pixels out of 264 in both the directions. To further improve the prediction on the boundaries, all overlapping patches are stacked together and then the final prediction is calculated as the average of each of the pixels. This is a commonly used approach for remote sensing problems (Audebert et al. 2016).

We start the training process for all the three networks of UNet, SegNet and UNet-AP from the scratch with using Adam optimizer with learning rate $\lambda = 0.001$ to 0.00001 decreasing by the factor of 10 and in total 200 epochs, which consists of iterative feeding of mini-batches to the network, computing the gradients and updating weights, until each patch in training data are processed once by the network. The layer depth of the network and the initial feature maps were improvised by values from 7 to 13 with increment factor of 2, insufficient layer depth lead to poor generalization capacity, while excessive layers vanish the backward gradient. The feature map intialised from 32 to 128 layers with the multiple of factor of two. The loss function, categorical cross entropy typically used for a

Table 1. Quantitative evaluation of the architectures on the data set.

Networks	Precision	Recall	IoU	F_1 measure	Accuracy (%)
UNet	0.60	0.85	0.54	0.70	89.6
UNet-AP	0.64	0.91	0.61	0.75	91.5
SegNet	0.55	0.89	0.52	0.68	85.2

single target class problem is used in our study. All the parameters were obtained empirically during investigation of the training process on the validation datasets.

7. Qualitative analysis

In order to quantitatively analyse and evaluate the experiment, we use the binary classification metrics precision, recall, intersection over union (IoU) and F_1 measure. Each building footprint polygon is either the true positive (TP), the true negative (TN), false positive (FP) and false negative (FN) (Goutte and Gaussier 2005). The metrics are defined as below

$$Accuracy = \frac{TP + TN}{TP + FN + TN + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$Precision = \frac{TP}{TP + FP}$$

$$IoU = \frac{TP}{TP + FP + FN}$$

$$F_1 measure = \frac{2 * Precision * Recall}{Precision + Recall}$$

Where, accuracy is the measure of degree of closeness of a predicted class to the actual class and precision is fraction of prediction positives which are actual positive, that is, the precision indicates how well a binary classifier avoids to classify a non-building as a building pixel. Recall indicates how well a binary classifier avoids missing building pixels. The F_1- score is equal to the harmonic mean of precision and recall.

8. Results and discussions

We employ models as described in Section 4 for semantic segmentation for building footprint extraction. The relative comparison of UNet, SegNet and UNet-AP networks for building mask generation on basis of urban settlement along with the merged Cartosat-2 series images and Ground truth is shown in (Figure 6). Comparison of these networks is done on test data of size of images 5000×5000 pixels. The UNet-AP yields 1.9% improvement over UNet and 6.3% that of SegNet. The evaluation over the test datasets of the proposed model attains 91.5% accuracy as shown in (Table 1). As per the urban settlement, UNet-AP model perform better for dense built-up area and isolated building with 96.7% and 83% accuracy respectively than slum areas, due to low spectral and textural variance among the buildings. Also, distances between buildings are less than 2 pixels which are difficult to delineate even through visual interpretation and resulting in creation of relatively poor training data for slum areas, with model accuracy of 72.5% (Table 2). F1-measure, IoU, Precision and Recall as per urban settlement, and on test data is shown in (Tables 1 and 2). Detailed boundaries of building are obtained from

Table 2. Quantitative evaluation of the architecture on the basis of urban settlement.

Building type	Precision	Recall	IOU	F_1 measure	Accuracy (%)
			UNet		
Dense built-up	0.61	0.83	0.58	0.70	95.2
Slum	0.54	0.64	0.43	0.60	67.7
Isolated	0.70	0.81	0.61	0.75	81.9
			SegNet		
Dense built-up	0.55	0.95	0.54	0.69	92.4
Slum	0.55	0.72	0.44	0.62	60
Isolated	0.69	0.83	0.60	0.75	82
			UNet-AP		
Dense built-up	0.66	0.94	0.64	0.77	96.7
Slum	0.56	0.77	0.49	0.65	72.5
Isolated	0.70	0.82	0.62	0.76	83

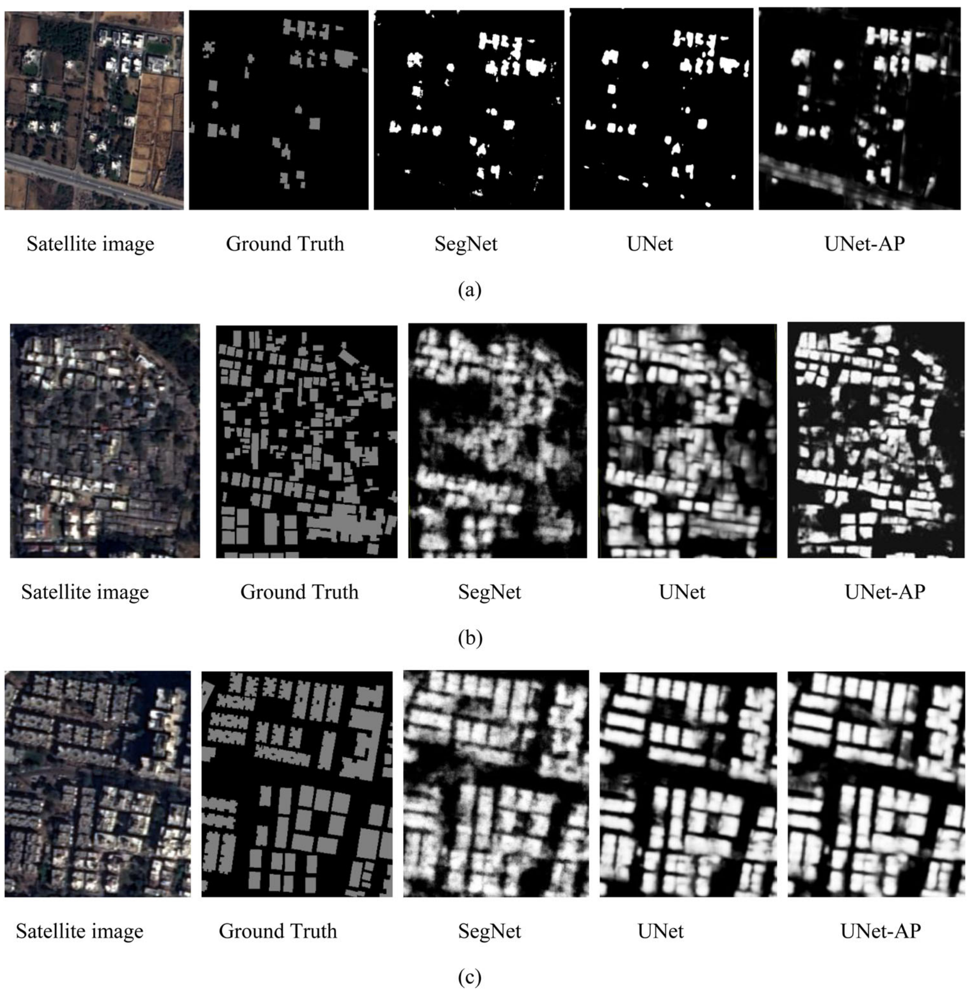


Figure 6. Relative performance of UNet, SegNet and UNet-AP networks for building mask generation on basis of urban settlement along with the merged Cartosat-2 series images and Ground truth (a) isolated building, (b) slum areas, (c) dense built-up.

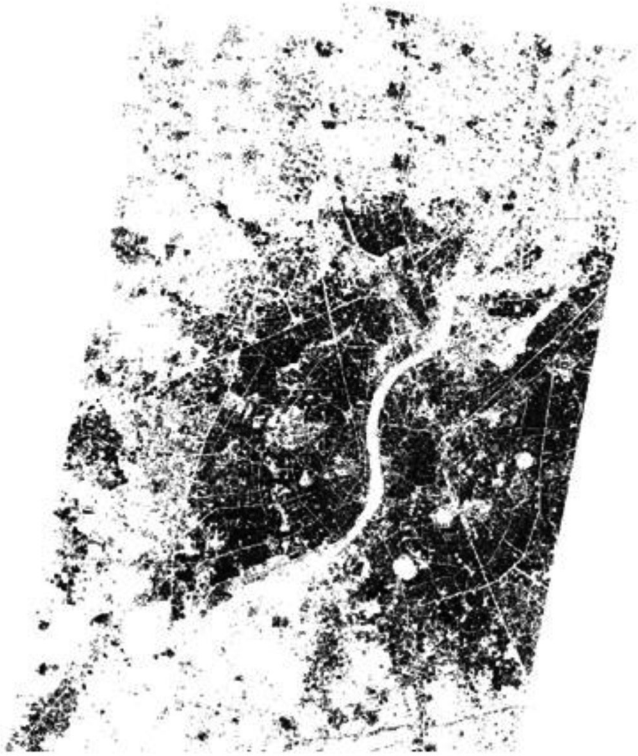


Figure 7. Building footprint map of the study area.

UNet-AP which lacks in UNet. ASPP encodes both object and image context at multiple levels, that produce semantically accurate prediction and detailed segmentation maps along the building boundaries. This could be easily seen in comparison of the predictions done on slum area in the test data. Even though it is visually very difficult to distinguish buildings, UNet-AP performed well and much better than other networks with better localization. In dense built-up all the network is comparable but building boundaries are sharply detected and no further post processing for correcting edges of the buildings is required for UNet-AP.

UNet-AP model substantially outperforms UNet and SegNet variants and is applied on the entire study area (Figure 7). The raster obtained a probability map with value in range (0,1) with 0 as the least probability for existence of the building and 1 with highest. The obtained raster results from the model for the entire study area is then converted into GIS format (polygons) for creating building maps. The building polygons with area less than $5\,m^2$ are removed, considering them erroneous polygons. The corrected building footprint obtained is shown in (Figure 7).

The proposed model can be applied to any other city with same terrain conditions such as planner areas for Ahmedabad city. However, for predicting building footprints in hilly terrain, since due to change in structures of building shape, size and characteristics more training data sets need to be provided along with existing training data, so that the model is able to generalize well across all terrain conditions.

As per classification of urban settlement, isolated building was considered as different class since these mainly found in and around classes such as fellow land, farm lands where the contextual information plays a vital role. The most difficult area of urban settlement was slum areas. These kinds of buildings have dark spectra and visually it is difficult to distinguish it from the other objects and buildings. Most of the buildings come under the class of dense built-up type of buildings and these buildings are correctly identified with high accuracy. Since most of the study area and training set is dominated by dense-built up, it shows good accuracy for this type of building. However, isolated buildings and slums are relatively less in number in training and study areas, resulting in relatively less accuracy. This can be substantially increased with increase in training samples for these kinds of buildings.

9. Conclusion

We proposed UNet-AP, a deep convolutional network architecture for semantic segmentation for extracting building footprint from very-high resolution remote sensing satellite imagery. The proposed network of UNet-AP by applying atrous convolution with up-sampled filters for dense feature extraction. ASPP, encodes objects as well as image context at multiple scales to produce semantically accurate predictions and detailed segmentation maps along the object boundaries. We analysed UNet-AP and compared it with other networks such as SegNet and UNet and found that prediction for building edges are highly improved and it results in better accuracy too. For future, further accuracy can be improved with the availability of high resolution and high-quality DEM for large areas and variants of terrain need to be included such as hilly terrain and coastal areas where the building structures have differences in their appearance and association of the buildings with nearby objects.

Disclosure statement

No potential conflict of interest was reported by the author(s).

References

[AMC] Ahmedabad Municipal Corporation. 2006. City Development Plan Ahmedabad 2006-2016. Ahmedabad Municipal Corporation, Ahmedabad.

Alshehhi R, Marpu PR, Woon WL, Mura MD. 2017. Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks. ISPRS J Photogramm Remote Sens. 130: 139–149.

Audebert N, Saux B, Le L. 2016. Semantic segmentation of earth observation data using multimodal and multi-scale deep networks. In Proceedings of Asian Conference on Computer Vision; Taipei, Taiwan; p. 180–196.

Audebert N, Saux B, Lefèvre S. 2018. Beyond RGB: very high resolution urban remote sensing with multimodal deep networks. ISPRS J Photogramm Remote Sens. 140:20–32.

Awrangjeb M, Ravanbakhsh M, Fraser CS. 2010. Automatic detection of residential buildings using LiDAR data and multispectral imagery. ISPRS J Photogramm Remote Sens. 65(5):457–467.

Badrinarayanan V, Kendall A, Cipolla R. 2017. SegNet: a deep convolutional encoder-decoder architecture for image segmentation. IEEE Trans Pattern Anal Mach Intell. 39(12):2481–2495.

Bittner K, Adam F, Cui S, Körner M, Reinartz P. 2018. Building footprint extraction from VHR remote sensing images combined with normalized DSMs using fused fully convolutional networks. IEEE J Sel Top Appl Earth Observ Remote Sens. 11(8):2615–2629.

Bottou L. 2010. Large-scale machine learning with stochastic gradient descent. In Proceedings of COMPSTAT'2010. Paris: Springer; p. 177–186.

Chen LC, Papandreou G, Kokkinos I, Murphy K, Yuille AL. 2018. Deeplab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. IEEE Trans Pattern Anal Mach Intell. 40(4):834–848.

Demir-Kavuk O, Riedesel H, Knapp EW. 2010. Exploring classification strategies with the CoEPrA 2006 contest. Bioinfomatics. 26(5):603–609.
Gavankar NL, Ghosh SK. 2018. Automatic building footprint extraction from high-resolution satellite image using mathematical morphology. Eur J Remote Sens. 51(1):182–193. No
Goutte C, Gaussier E. 2005. A Probabilistic interpretation of precision, recall and F-score, with implication of evaluation. In: Losada DE, Fernandez-Luna JM, editors. Advances in information retrieval. ECIR 2005. Lecture notes in computer sciences. Berlin, Heidelberg: Springer; p. 3408.
Guercke R, Sester M. 2011. Building footprint simplification based on Hough transform and least squares adjustment. In Proceedings of 14th Workshop ICA Commission Generalisation Multiple Represent: Paris, France; p. 1–9.
Haklay M, Weber P. 2008. OpenStreetMap: user-generated street maps. IEEE Pervasive Comput. 7(4): 12–18.
Huertas A, Nevatia R. 1988. Detecting buildings in aerial images. ComputVis Graph Image Process. 41(2): 131–152.
Irvin RB, McKeown DM. 1989. Methods for exploiting the relationship between buildings and their shadows in aerial imagery. IEEE Trans Syst Man Cybern. 19(6):1564–1575.
Jensen JR, Cowen DC. 1999. Remote sensing of urban suburban infrastructure and socio-economic attributes. Photogramm Eng Remote Sens. 65(5):611–622.
Kim T, Muller JP. 1999. Development of a graph-based approach for building detection. Image Vis Comput. 17(1):3–14.
Li X, Shao G. 2014. Object based land cover mapping with high resolution arial photography at a county scale in Midwestern USA. Remote Sens. 6(11):11372–11390.
Lnglada J. 2007. Automatic recognition of manmade objects in high resolution optical remote sensing images by svm classification of geometric image features. ISPRS J Photogramm Remote Sens. 62(3): 236–248.
Long J, Shelhamer E, Darrell T. 2015. Fully convolutional networks for semantic segmentation. Proceedings of the IEEE; Boston: University of Berkeley.
Prathap G, Afanasyev I. 2018. Deep learning for building detection in satellite multispectral imagery. 2018 International Conference on Intelligent System (IS). IEEE: Funchal - Madeira, Portugal; p. 461–465. DOI:10.1109/IS.2018.8710471.
Ronneberger O, Fischer P, Brox T. 2015. UNet: convolutional networks for biomedical image segmentation. Germany: Computer Science Department and BIOSS Centre for Biological Signalling Studies, University of Freiburg.
Simonyan K, Zisserman A. 2014. Very deep convolutional networks for large-scale image recognition.*CoRR, abs/1409.1556*.
Tan Y, Yu Y, Xiong S, Tian J. 2016. Semiautomatic building extraction from very high resolution remote sensing imagery via energy minimization model. In 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS). Beijing: IEEE; p. 657–660.
Wang O, Lodha SK, Helmbold DP. 2006. A Bayesian approach to building footprint extraction from aerial LIDAR data. In Third International Symposium on 3D Data Processing, Visualization and Transmission (3DPVT06). Chapel Hill (NC): IEEE; p. 192–199.
Wu G, Shao X, Guo Z, Chen Q, Yuan W, Shi X, Xu Y, Shibasaki R. 2018. Automatic building segmentation of aerial imagery using multi-constraint fully convolutional networks. Remote Sens. 10(3):407.
Xu Y, Wu L, Xie Z, Chen Z. 2018. Building extraction in very high resolution remote sensing imagery using deep learning and guided filters. Remote Sens. 10(1):144.
Xu Y, Xie Z, Feng Y, Chen Z. 2018. Building extraction in very high resolution remote sensing imagery using deep learning and guided filters. Remote Sens. 10(9):1461.
Yang HL, Yuan J, Lunga D, Laverdiere M, Rose A, Bhaduri B. 2018. Building extraction at scale using convolutional neural network: mapping of the United States. IEEE J Sel Top Appl Earth Observ Remote Sens. 11(8):2600–2614.
Yuan J. 2018. Learning building extraction in aerial scenes with convolutional networks. IEEE Trans Pattern Anal Mach Intell. 40(11):2793–2798.
Zhang Q, Wang Y, Liu Q, Liu X, Wang W. 2016. CNN based suburban building detection using monocular high resolution google earth images. 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS). Beijing (China): IEEE. p. 661–664.
Zhou QY, Neumann U. 2008. Fast and extensible building modelling from airborne LiDAR data. In Proceedings of 16th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems; Irvine California.